

SafeOT-Dual: one price rule, two views

Penalty view

$$\max_{\pi} J_r(\pi) - \lambda^T J_c(\pi)$$

Exact settings

$$c = 0$$

Fixed β

fixed penalty
 $\lambda = \beta$

Adaptive β

Lagrangian feedback

Joint constraints

$$C_1, b_1$$

$$C_2, b_2$$

...

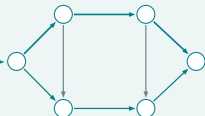
$$C_K, b_K$$

Hard support H



$$F_H = 0$$

One graph problem



$$\min_F - \langle A, F \rangle + \varepsilon KL(F \| \hat{F})$$

flow balance

$$\langle C_k, F \rangle \leq b_k$$

Shared safety price

$$\lambda^*$$

Clip + filter

$$\lambda_k = \lambda_k^* + \beta_k$$

graph price

feedback

$$\lambda_1$$

$$\lambda_2$$

...

$$\lambda_K$$

Constraint view

$$\max_{\pi} J_r(\pi) \text{ s.t. } J_c(\pi) \leq B$$

Graph-price limits

under assumptions

$$\varepsilon \rightarrow 0$$

LP-derived price

$$\varepsilon \rightarrow \infty$$

clipped active-set price

One policy-update rule

$$\tilde{A} = (A_r - \sum_k \lambda_k A_{c_k}) / (1 + \sum_k \lambda_k)$$



PPO / TRPO